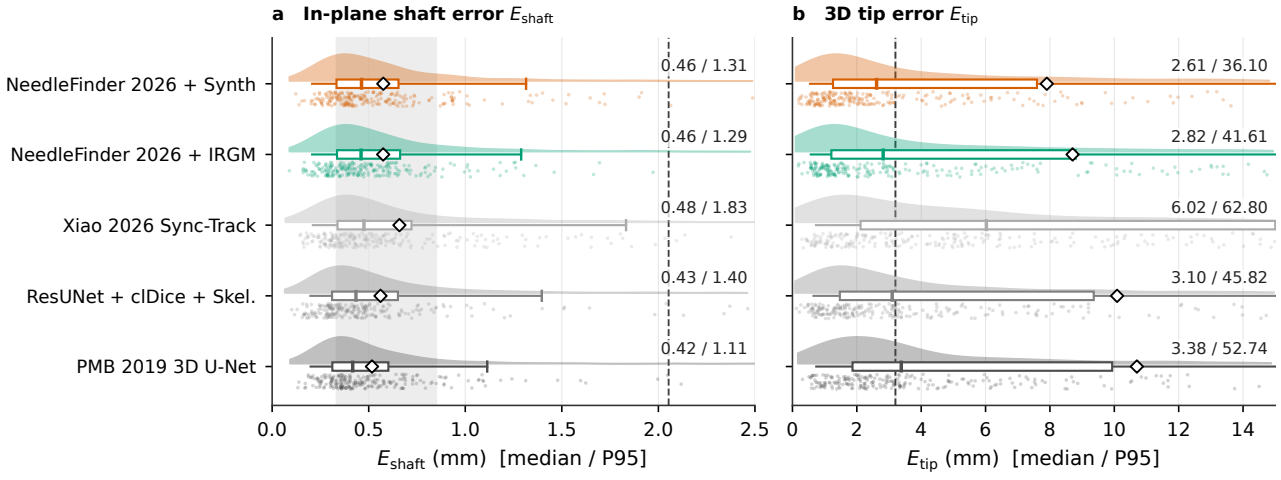

a In-plane shaft error E_{shaft}
b 3D tip error E_{tip}
NeedleFinder 2026 + Synth
NeedleFinder 2026 + IRGM
Xiao 2026 Sync-Track
ResUNet + clDice + Skel.
PMB 2019 3D U-Net
0.46 / 1.31
0.46 / 1.29
0.48 / 1.83
0.43 / 1.40
0.42 / 1.11
2.61 / 36.10
2.82 / 41.61
6.02 / 62.80
3.10 / 45.82
3.38 / 52.74
0.0
0.5
1.0
1.5
2.0
2.5
0
2
4
6
8
10
12
14
E_{shaft} (mm) [median / P95]
E_{tip} (mm) [median / P95]